

**Probabilidades e Estatística**  
LEI, LEIRT, LIG

**Formulário Teste 2 e FF**

|                                                                                                                                                                        |                                                                                                                                                          |                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{\sum_{i=1}^n x_i^2 - n\bar{x}^2}{n-1}$                                                                     | $r = \frac{1}{n-1} \sum_{i=1}^n \frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y}$                                                                     | $r_1 = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$                                             |
| $y_i = a + bx_i + e_i$                                                                                                                                                 | $\hat{a} = \bar{y} - \hat{b}\bar{x}$                                                                                                                     | $\hat{b} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$      |
| $r^2 = \frac{\left(\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}\right)^2}{\left(\sum_{i=1}^n x_i^2 - n\bar{x}^2\right) \times \left(\sum_{i=1}^n y_i^2 - n\bar{y}^2\right)}$ |                                                                                                                                                          |                                                                                                 |
| $P(A B) = \frac{P(A \cap B)}{P(B)}$                                                                                                                                    | $P(\overline{A}) = 1 - P(A)$                                                                                                                             | $P(B \setminus A) = P(B) - P(A \cap B)$                                                         |
|                                                                                                                                                                        | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$                                                                                                                |                                                                                                 |
| $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$                                                               |                                                                                                                                                          |                                                                                                 |
| $P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2 A_1)P(A_3 A_1 \cap A_2) \dots P(A_n A_1 \cap A_2 \cap \dots A_{n-1})$                                               |                                                                                                                                                          |                                                                                                 |
| $P(B) = P(B A_1)P(A_1) + \dots + P(B A_m)P(A_m)$                                                                                                                       |                                                                                                                                                          |                                                                                                 |
| $P(A_i B) = \frac{P(B A_i)P(A_i)}{P(B)} = \frac{P(B A_i)P(A_i)}{P(B A_1)P(A_1) + \dots + P(B A_m)P(A_m)}$                                                              |                                                                                                                                                          |                                                                                                 |
| $P(x_0 < X \leq x_1) = F_X(x_1) - F_X(x_0)$                                                                                                                            | $E(X) = \sum_x x f_X(x)$                                                                                                                                 | $p \leq F_X(q_p) \leq p + P(X = q_p)$                                                           |
| $E(aX + b) = aE(X) + b$                                                                                                                                                | $V(X) = E(X^2) - E^2(X)$                                                                                                                                 | $V(aX + b) = a^2V(X)$                                                                           |
| $X \sim \text{Bernoulli}(p)$<br>$P(X = x) = p^x(1 - p)^{1-x}, \; x = 0, 1$<br>$E(X) = p \quad \text{e} \quad V(X) = p(1 - p)$                                          | $X \sim \text{Binomial}(n, p)$<br>$P(X = x) = \binom{n}{x} p^x(1 - p)^{n-x}, \; x = 0, 1, \dots, n$<br>$E(X) = np \quad \text{e} \quad V(X) = np(1 - p)$ |                                                                                                 |
| $X \sim N(\mu, \sigma^2)$<br>$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$                                                                                                 | $\frac{S_n - E(S_n)}{\sqrt{V(S_n)}} \overset{a}{\sim} N(0, 1)$                                                                                           | $\frac{\overline{X}_n - E(\overline{X}_n)}{\sqrt{V(\overline{X}_n)}} \overset{a}{\sim} N(0, 1)$ |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                |                                                                  |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------------------------|------------------------------|------------------------------------------------|------------------------------------------------------------------|---------------------------|-------------------------------------|-----------------------------------|---------------------------|-------------------------------------|-----------------------------------|
| $P(a < Z < b) = \Phi(b) - \Phi(a)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\Phi(-x) = 1 - \Phi(x)$                       | $\Phi^{-1}(p) = -\Phi^{-1}(1 - p)$                               |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
| $X \sim N(\mu, \sigma^2) \quad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad \mathbf{z.test}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                |                                                                  |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
| $X \sim N(\mu, \sigma^2) \quad T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)} \quad \mathbf{t.test}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                |                                                                  |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
| $X \sim N(\mu, \sigma^2) \quad Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \quad \mathbf{variance.test}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                |                                                                  |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
| $X \sim \text{Bernoulli}(p) \quad Z = \frac{\bar{X} - p}{\sqrt{\frac{\bar{X}(1-\bar{X})}{n}}} \overset{a}{\sim} N(0, 1) \quad \frac{\bar{X} - p}{\sqrt{\frac{p(1-p)}{n}}} \overset{a}{\sim} N(0, 1) \quad \mathbf{prop.test/binom.test}$                                                                                                                                                                                                                                                                                                                                                                           |                                                |                                                                  |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
| <table border="1"> <tr> <td><math>H_1 : \theta \neq \theta_0</math></td><td><math>RC = \{t \in \mathbb{R} : t &lt; a \vee t &gt; b\}</math></td><td><math>\mathbf{valor-p} = 2 \times \min\{P(T_0 &lt; t_0), P(T_0 &gt; t_0)\}</math></td></tr> <tr> <td><math>H_1 : \theta &gt; \theta_0</math></td><td><math>RC = \{t \in \mathbb{R} : t &gt; b\}</math></td><td><math>\mathbf{valor-p} = P(T_0 &gt; t_0)</math></td></tr> <tr> <td><math>H_1 : \theta &lt; \theta_0</math></td><td><math>RC = \{t \in \mathbb{R} : t &lt; a\}</math></td><td><math>\mathbf{valor-p} = P(T_0 &lt; t_0)</math></td></tr> </table> |                                                |                                                                  | $H_1 : \theta \neq \theta_0$ | $RC = \{t \in \mathbb{R} : t < a \vee t > b\}$ | $\mathbf{valor-p} = 2 \times \min\{P(T_0 < t_0), P(T_0 > t_0)\}$ | $H_1 : \theta > \theta_0$ | $RC = \{t \in \mathbb{R} : t > b\}$ | $\mathbf{valor-p} = P(T_0 > t_0)$ | $H_1 : \theta < \theta_0$ | $RC = \{t \in \mathbb{R} : t < a\}$ | $\mathbf{valor-p} = P(T_0 < t_0)$ |
| $H_1 : \theta \neq \theta_0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $RC = \{t \in \mathbb{R} : t < a \vee t > b\}$ | $\mathbf{valor-p} = 2 \times \min\{P(T_0 < t_0), P(T_0 > t_0)\}$ |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
| $H_1 : \theta > \theta_0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $RC = \{t \in \mathbb{R} : t > b\}$            | $\mathbf{valor-p} = P(T_0 > t_0)$                                |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |
| $H_1 : \theta < \theta_0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $RC = \{t \in \mathbb{R} : t < a\}$            | $\mathbf{valor-p} = P(T_0 < t_0)$                                |                              |                                                |                                                                  |                           |                                     |                                   |                           |                                     |                                   |